

Georges Birangou

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Casablanca, Morocco

PROFILE

Data & AI Engineer with hands-on experience in ETL pipeline design, cloud data migration, and machine learning systems. Skilled in Python, Spark, Kafka, and Azure, with a track record of delivering end-to-end data solutions, from raw ingestion to production-ready insights. Passionate about building scalable, impact-driven architectures at the intersection of Big Data and AI.

PROFESSIONAL EXPERIENCE

Data & AI Engineer Intern

Mar. 2026 – Sep. 2026

Cetim Maroc Développement

Casablanca, Morocco

- Built an end-to-end Python system to automate HCF test report generation, fully replacing existing VBA macros and eliminating manual reporting overhead
- Designed a multi-source ETL pipeline extracting data from heterogeneous Excel files, Azure data lake and raw machine files, normalizing and loading into MongoDB
- Developed an agentic column mapping module powered by a local LLM (Qwen via Ollama), autonomously inferring and aligning heterogeneous data schemas with zero cloud dependency
- Generated COFRAC-certified PDF reports (ReportLab) with a PyQt6 desktop interface featuring real-time preview
- Packaged the application as a standalone EXE via PyInstaller for zero-dependency deployment on end-user workstations

Software Engineer Intern

Jun. 2025 – Sep. 2025

Chambre de Commerce de Casablanca-Settat

Casablanca, Morocco

- Designed a cross-platform library management system in React Native (mobile) and Next.js (web), both connected to a shared Supabase backend via REST APIs, managing a catalog of **500+** books with real-time synchronization
- Implemented book catalog browsing, borrow/return tracking, full-text search, and automated notifications, **significantly reducing** administrative overhead by **saving 1.5 hour/week** for library staff
- Built an analytics dashboard tracking key library KPIs (most-read books, peak attendance days, and weekly reservation volume), giving staff actionable insights at a glance

PROJECTS

Real-Time Banking Fraud Detection System | *Python, XGBoost, Kafka, PostgreSQL, Grafana, Docker* 2025

- Trained an XGBoost + SMOTE model on **284,807** transactions to detect fraud in a highly imbalanced dataset (**0.17%** fraud rate), achieving ROC-AUC of **0.978**
- Reached **89.8%** recall on the test set, intercepting **88 out of 98** fraudulent transactions while keeping false alerts at **0.3%**
- Built a real-time Kafka streaming pipeline for transaction ingestion and ML inference, with results stored in PostgreSQL and monitored via a Grafana supervision dashboard
- Deployed the full architecture in Docker for containerized, reproducible environments

IoT Industrial Supervision Architecture | *Python, Kafka, Spark, TimescaleDB, MinIO, scikit-learn* 2025

- Ingested real-time sensor streams (temperature, vibration) via Apache Kafka and processed them with Spark Streaming, applying type-aware validation filters before storage
- Persisted raw data to MinIO (S3-compatible) and computed **1-minute sliding window KPIs** into TimescaleDB via JDBC, with **30-second watermarking** for late-arrival tolerance
- Trained a RandomForest model on **1,084 temperature readings** to predict values **5 time steps ahead** using a rolling window of 10 measurements, with live inference fed from Kafka
- Implemented an autonomous safety trigger that automatically shuts down the sensor simulator upon detecting critical temperature thresholds

TECHNICAL SKILLS

Languages: Python (Pandas, NumPy, Scikit-learn), SQL, TypeScript/JavaScript, HTML/CSS
Frameworks: React Native, React JS, Next.js, Django
Big Data & AI: Apache Kafka, Apache Spark, Machine Learning, Deep Learning, Generative AI, RAG
Cloud & Databases: Azure, PostgreSQL, MySQL, TimescaleDB, MinIO, Supabase, SQLite
Visualization: Grafana, Power BI, Apache Superset
DevOps & Tools: Docker, Kubernetes, Git/GitHub, Hugging Face, ClearML

EDUCATION

EIGSI Casablanca – School of Industrial Systems Engineering	Casablanca, Morocco
<i>Master’s-level Engineering Degree – Specialization: Artificial Intelligence & Big Data</i>	<i>2022 – 2026</i>

CERTIFICATIONS

Oracle Cloud Infrastructure 2025 Generative AI Professional – Oracle
Oracle Cloud Infrastructure Data Science Professional – Oracle

LANGUAGES

French: Advanced (C1) | **English:** Intermediate (B2)